

Suraj Patel

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SUMMARY

Dynamic Software Engineer specializing in agentic AI workflows and multimodal RAG platforms, with a proven track record in developing real-time backend systems. Expert in building production-level LLM applications utilizing Python, FastAPI, Node.js, and LangChain, with a strong focus on scalable REST APIs, Redis caching, and cloud deployment. Committed to delivering innovative solutions that enhance performance and user experience.

EXPERIENCE

Careerboat

Dec 2025 - Present

Full Stack Engineer,

- Reduced LLM token consumption by 40% for the AI interview service by architecting stateful Node.js APIs with MongoDB checkpointing and dynamic context summarization across multi-turn sessions, significantly enhancing efficiency.
- Engineered a real-time streaming layer over Server-Sent Events and AWS S3, achieving a time-to-first-audio of approximately 2 seconds, which improved user engagement by delivering token-by-token responses.
- Revamped a high-volume job search API by redesigning MySQL FULLTEXT queries and ranking algorithms, slashing search latency from 1-5 minutes to just 5-10 seconds while enhancing search relevance.
- Integrated the Razorpay payment gateway for secure transactions, implementing custom backend logic for discount coupons, which streamlined the payment process.

EDUCATION

Amity University Online

Jul 2024 - Jun 2026

Master's in Master of Computer Applications (MCA)

Lucknow Christian College

Jul 2021 - Jun 2024

Bachelor's in Bachelor of Science

PROJECTS

InsightFlow – Multimodal RAG Platform

insightflow.surajpatel.dev | github.com/SurajPatel04/ai-multimedia-rag-app

- Designed a two-phase ingestion flow that ensures uploads are confirmed before processing, effectively eliminating unnecessary embedding costs on unconfirmed files.
- Integrated Deepgram for timestamped transcription with clickable media citations, facilitating precise retrieval and playback of media content to the exact second.
- Implemented Redis semantic caching for repeated queries, alongside LangGraph-based conversation memory management, and deployed on GCP using Docker and GitHub Actions CI/CD.

AI Manim Video Generator

video.surajpatel.dev | github.com/SurajPatel04/manimVideoGenerate

- Engineered a self-healing generation loop with Vision QA, leveraging Vision LLMs (GPT-4o-mini/Gemini) to visually validate rendered frames against user intent, boosting execution success rates from approximately 60% to over 90%.

- Developed a scalable, asynchronous rendering architecture utilizing Celery, Redis, and Supabase to support concurrent generation, real-time polling, and in-flight task cancellation, secured through JWT and Google OAuth.

SKILLS

- **Languages:** JavaScript, TypeScript, Python
- **Frontend:** React, Redux, HTML5, CSS3
- **Backend:** Node.js, Express.js, FastAPI, REST APIs, JWT, OAuth 2.0
- **Databases & Caching:** PostgreSQL (Prisma), SQL, MongoDB (Mongoose, Beanie), Redis, FAISS (Vector DB)
- **AI:** LangChain, LangGraph, LangSmith (LLM observability), OpenAI API, Gemini API, RAG, Prompt Engineering
- **Tools & DevOps:** AWS (S3), GCP, Docker, Git, GitHub Actions, Linux, Bash/Shell, Celery
- **Core Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming (OOP), System Design